ISA/IEC 62443-3-3 / 4-2 • SL-3 TESTING CHEAT SHEET

ProSoft MVI56E-MCM / MCMXT Modbus Master/Slave Module (in-chassis, ControlLogix) · Component type: **Embedded Device (EDR) - in-chassis serial Modbus master/slave communication module** · Target SL = SL 3

Where to work	In-chassis 1756 ControlLogix module **owned by the ControlLogix processor** (data exchanged over the backplane via a **Studio 5000 Add-On Instruction (AOI)** + controller tags). Configure with **ProSoft Configuration Builder (PCB)** over the module's **config/diagnostics Ethernet port** (or via **CIPconnect** through a 1756-EN2T/EN4TR) - **ProSoft Discovery Service (PDS)** assigns a temp IP; an on-board **web server** serves module docs/sample. Modbus is **SERIAL**: two ports (RS-232/422/485), each Modbus RTU/ASCII **Master and/or Slave** (up to 325 commands/port). Harden = disable/segment the config port + web after commissioning; the ControlLogix + **FactoryTalk Security** govern the processor side.
Ports / paths	**Modbus RTU/ASCII** on the two **SERIAL** ports (RS-232/422/485) - Master and/or Slave. The **10/100 Ethernet port = config + diagnostics ONLY** (PDS discovery, PCB config, on-board **HTTP web**, CIPconnect bridging) - **NOT Modbus TCP**. **1756 backplane** (AOI data exchange). 4-char LED status display. No TLS/crypto anywhere.
Notes / lifecycle	In-chassis MVI56E-MCM / MCMXT (XT = extended temperature). No user accounts, no comms encryption. Native controls = **disable/segment the config Ethernet + web**, protect the PCB **.ppf** config, the owning **ControlLogix + FactoryTalk Security / keyswitch**, and physical control; the serial Modbus is contained by the RS-485 loop. Confirm module firmware (rev 3 = 10,000 registers); track ProSoft / CISA advisories.
How to read	Default read-only (PCB read-only config review; read-only serial Modbus reads; module LED/web diagnostics; Studio 5000 AOI/tag review). Each row = native step-by-step + command-level external-tool validation + pass. Write/exploit tools = isolated bench + written authorization. Several rows are Compensating / N-A-at-device.

FR 1 — Identification & Authentication Control (IAC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 1.1	Human user identification & auth	The module has **no user accounts**. Two access surfaces: (1) the **config/diagnostics Ethernet port** - a host on that network can reconfigure the module via **PCB** (PDS discovery + the on-board web have minimal/no auth), so **disable/segment the config port** after commissioning; (2) the **processor side** (backplane data) is governed by the **ControlLogix + FactoryTalk Security** (who may go online / download the AOI/tags). **Test:** from the config network, run PDS / PCB Connect -> confirm the port is segmented/disabled off-commissioning; browse the module web -> docs only, no config exposure. **Ext. tools:** PDS / PCB (discover/connect from the config net); `curl -i http:///`; FactoryTalk Security (processor-side policy).	Config port disabled/segmented; processor-side authorized by FactoryTalk; no anonymous reconfig.	EDR - no module accounts; compensating.
SR/CR 1.2	Device & software-process I&A;	Serial Modbus is **unauthenticated**; the module (as slave) is identified by its **Modbus station address** per port, and the config port by IP. Bind identity by a single known master + the physical serial loop + segmentation of the config port. **Test:** from a non-authorized master on the serial loop, poll the module -> blocked by loop control. **Ext. tools:** `modpoll -m rtu -a -b -1 /dev/ttyUSB0` (read-only, authorized); confirm serial station address + config-port IP.	Module identified by station address/IP; single authorized master; unauthenticated Modbus compensated.	No device auth - Compensating.
SR/CR 1.3	Account management	No module accounts; manage config-network access, the PCB **.ppf** custody, and **FactoryTalk Security** on the owning processor (least privilege, no shared logins). **Ext. tools:** Config-net access review; .ppf custody record; FactoryTalk Security (user/role).	Config access + .ppf + processor identities controlled; defaults removed.	
SR/CR 1.4	Identifier management	Confirm a unique **Modbus station address** per serial port (as slave), a unique config-port IP, and a unique module identity (slot + serial); no reuse. **Ext. tools:** PCB (address/IP); `modpoll` scan for collisions; Studio 5000 (module identity); asset register.	Unique serial address + config IP + module identity; no reuse.	
SR/CR 1.5	Authenticator management	The module stores **no credentials** and the config port/web have no strong auth - compensate by disabling/segmenting the config port; enforce **FactoryTalk Security** password policy on the owning processor. **Ext. tools:** Confirm config port disabled/segmented; FactoryTalk password policy.	No module credentials (config port locked down); processor policy enforced.	Config-port lockdown.

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR 1.6	Wireless access management	**N/A.** Wired in-chassis module with no wireless interface. **Ext. tools:** Confirm no wireless.	N/A - justification recorded (wired device).	N/A
SR/CR 1.7	Strength of password-based authentication	Confirm the password-strength policy (length/complexity/lifetime) meets SL-3. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.5** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of password-based authentication verified individually.	
SR/CR 1.8	Public key infrastructure (PKI) certificates	**N/A at device.** No PKI/TLS/crypto (serial Modbus + HTTP config). Record N/A; integrity/confidentiality by segmentation + the RS-485 loop. **Ext. tools:** Confirm no certificate/crypto; serial + config capture (cleartext).	N/A device PKI; compensating segmentation.	N/A.
SR/CR 1.9	Strength of public key-based authentication	Confirm public-key authentication uses strong, current keys/algorithms. **Steps:** for this product this is assessed with the same evidence as **SR/CR 1.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Strength of public key-based authentication verified individually.	
SR/CR 1.10	Authenticator feedback	**N/A at device** (no module login). Applies to FactoryTalk on the processor. **Ext. tools:** FactoryTalk visual verification.	N/A at device; FactoryTalk masks entry.	N/A.
SR/CR 1.11	Unsuccessful login attempts	**N/A at device** (no login). Remove exposure with config-port lockdown + segmentation; FactoryTalk lockout on the processor; monitor anomalous serial/config activity. **Ext. tools:** FactoryTalk lockout; serial/OT IDS.	No device login; config lockdown + processor lockout + monitoring.	Compensating.
SR/CR 1.12	System use notification	**N/A at device.** Placard on the cabinet; banner on the Studio 5000 EWS (compensating). **Ext. tools:** Physical placard; EWS banner.	Use notice via placard/EWS (compensating).	Compensating.
SR 1.13	Access via untrusted networks	The **config Ethernet port must not be on an untrusted/routable network** - segment it (or disable it after commissioning); the serial Modbus is local. No untrusted access to the config port, the web, or the serial loop; CIPconnect config via a controlled path only. **Test:** from an untrusted VLAN, reach the config port / web -> must fail. **Ext. tools:** `nmap -Pn -p 80` from the untrusted side (expect filtered); confirm no rogue bridge on the serial loop; firewall review.	Config port + serial loop unreachable from untrusted networks; config via a controlled path.	

FR 2 — Use Control (UC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 2.1	Authorization enforcement	No module roles. Authorization is (a) **disable/segment the config port** so the .ppf cannot be changed, and (b) the **ControlLogix processor + FactoryTalk Security / keyswitch** govern the backplane data (the AOI/tags). Scope which Modbus registers/commands are writable in the PCB **command list / slave map**. **Test:** with the config port locked, attempt a PCB Download -> blocked; review the command list for unintended write commands. **Ext. tools:** PCB (command-list write scope); confirm config-port lockdown; Studio 5000/FactoryTalk (processor authz).	Config change gated by port lockdown + processor FactoryTalk; write commands scoped.	No device roles - Compensating.
SR/CR 2.2	Wireless Use Control	**N/A.** No wireless interface on this device; wireless access is managed at the network layer.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 2.3	Use control for portable/mobile devices	Control the PCB laptop + the **.ppf** config file; only authorized firmware loaded; the module is in a chassis - control physical access. **Ext. tools:** Host/media control; .ppf ACL; physical inspection (chassis).	Engineering media + config file controlled.	
SR/EDR 2.4	Mobile code	**N/A.** Fixed-function module firmware executing the compiled command map - no general-purpose or downloadable code. **Ext. tools:** Confirm fixed-function device (no code path beyond firmware).	N/A - no mobile-code capability (fixed firmware).	EDR-specific / N/A.
SR/CR 2.5	Session lock	**N/A at device** (no interactive session). EWS OS lock for PCB. **Ext. tools:** EWS inactivity-lock GPO.	N/A at device; EWS locks.	N/A.

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 2.6	Remote session termination	Serial is single-master per port; the module bounds concurrent Modbus (per-port) and PCB diagnostic sessions; confirm no conflicting concurrent config. **Ext. tools:** PCB (diagnostics/connections); serial capture (one master per port).	Serial concurrency bounded per port; config session controlled.	
SR/CR 2.7	Concurrent session control	Confirm concurrent sessions are bounded/limited and monitored. **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Concurrent session control verified individually.	
SR/CR 2.8	Auditable events	The module has **no native security log**. Capture change attribution at the **ControlLogix processor + FactoryTalk AssetCentre** (AOI/tag + module-config changes), the PCB **.ppf version**, and a serial/Modbus-aware **OT IDS**; forward to the SIEM. **Ext. tools:** AssetCentre (processor/module change); PCB .ppf version control; serial/Modbus IDS + SIEM.	Changes attributable at processor/AssetCentre + .ppf version; Modbus events captured by IDS.	No native log - Compensating.
SR/CR 2.9	Audit storage capacity	Logging is off-module; confirm the processor/AssetCentre/SIEM are sized/retained and alert on collection or comms-loss failure. **Ext. tools:** AssetCentre/SIEM retention + comms-loss alert.	Off-module logs sized/retained; audit-failure/comms-loss alerting defined.	Compensating.
SR/CR 2.10	Response to audit processing failures	Confirm a defined response to audit-processing/forwarding failures (alerting; no silent loss). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.9** above (shared steps); confirm this requirement individually and adjudicate it separately.	Response to audit processing failures verified individually.	
SR/CR 2.11	Timestamps	The module has no RTC; events are timestamped by the ControlLogix processor / SCADA - confirm those are time-synced (CIP time / NTP) so module events correlate. **Ext. tools:** Processor WallClockTime / SCADA NTP check.	Correlatable time via the synced processor/SCADA.	
SR/CR 2.12	Non-repudiation	Confirm actions are attributable to a specific user (non-repudiation). **Steps:** for this product this is assessed with the same evidence as **SR/CR 2.8** above (shared steps); confirm this requirement individually and adjudicate it separately.	Non-repudiation verified individually.	
EDR 2.13	Physical diagnostic & test interfaces	Control physical interfaces: the module in a **locked chassis/cabinet**; the **config Ethernet port** (disable after commissioning), the **serial ports**, and the backplane; secure the config-network drop. **Ext. tools:** Physical inspection (chassis lock, config-port drop, serial terminals); confirm config port disabled.	Physical/diagnostic interfaces controlled; config port disabled post-commissioning.	EDR-specific.

FR 3 — System Integrity (SI)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 3.1	Communication integrity	Serial Modbus and the config Ethernet (PCB/web) are **unauthenticated cleartext** (no TLS) - integrity/confidentiality come from **containment**: keep the serial loop local and **disable/segment the config port + web**; no cleartext config crosses a zone boundary. **Test:** confirm serial Modbus stays on its loop and the config port is off/segmented. **Ext. tools:** Serial capture / `modpoll` (cleartext); `nmap -p 80` (config web off/segmented).	Serial + config comms contained/segmented; no uncontrolled cleartext.	No crypto - Compensating.
SR/EDR 3.2	Protection from malicious code (component)	Confirm only ProSoft firmware runs (fixed-function). Load firmware only from ProSoft via PCB and verify provenance; restrict who can update (config-port lockdown); track ProSoft / CISA advisories. **Ext. tools:** PCB (firmware version + provenance); ProSoft advisory review.	Only vendor firmware runs; provenance verified; update restricted.	EDR-specific.
SR/CR 3.3	Security functionality verification	After any change re-verify: config port disabled/segmented, the command/register map, serial settings, firmware, and processor-side FactoryTalk, against the baseline; compare the PCB **.ppf**. **Ext. tools:** PCB (config vs baseline .ppf); `nmap -p 80` ; FactoryTalk review.	Security settings match the baseline after change.	
SR/CR 3.4	Software & information integrity	Protect the PCB **.ppf** config and the **AOI/controller tags** in the Studio 5000 project: host ACLs, version control; hash the .ppf; the ControlLogix keyswitch/FactoryTalk prevent processor-side change; detect drift by compare. **Ext. tools:** Host ACL; `certutil -hashfile .ppf SHA256`; Studio 5000 online-vs-offline compare.	Config + AOI integrity protected; unauthorized change prevented/detected.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 3.5	Input validation	Confirm the module tolerates malformed **Modbus / config** frames without faulting or corrupting the map. **ISOLATED bench only:** fuzz the serial Modbus + the config port on a bench module and watch the LED/AOI status. **Ext. tools:** Serial/Modbus fuzzer (`boofuzz`) on a bench module; watch LED status + AOI status word.	No fault or map corruption on malformed input.	Do NOT fuzz an in-service module.
SR/CR 3.6	Deterministic output	Verify **comm-fail behavior**: on serial Modbus loss the module signals status to the AOI, and the **ControlLogix logic** drives the process to a defined state (a stale Modbus read must not present as live). Confirm the command-error / comm-fail handling. **Ext. tools:** PCB (comm-fail/command-error config); Studio 5000 (AOI status -> fault logic); simulate serial loss.	Comm loss signaled to the processor; process assumes a defined state (no stale data as live).	
SR/CR 3.7	Error handling	Confirm graceful error handling: the module **LED display** + the AOI status/error words report faults without unsafe behavior or info leak. **Ext. tools:** Module LED review; Studio 5000 (AOI status/error words).	Graceful error handling; no unsafe state or info leak.	
SR/CR 3.8	Session integrity	**N/A / limited.** No authenticated/encrypted session; integrity by containment. Record N/A + compensating segmentation. **Ext. tools:** Confirm no session security; segmentation review.	N/A/limited at device; compensating segmentation recorded.	N/A/Partial.
SR/CR 3.9	Protection of audit information	Since audit is off-module, protect the AssetCentre / IDS / SIEM store and restrict who can read/clear it. **Ext. tools:** AssetCentre / SIEM access review.	Off-module audit records protected from unauthorized access.	
EDR 3.10	Support for updates	Update firmware via **PCB** with firmware from ProSoft; verify version/provenance; plan in a maintenance window with a config (.ppf) restore/rollback plan; re-lock the config port after. **Ext. tools:** PCB (update flow + provenance); ProSoft firmware; rollback/.ppf restore plan.	Controlled updates from a verified source, with rollback; config port re-locked.	EDR-specific.
EDR 3.11	Physical tamper resistance/detection	Keep the chassis in a locked/sealed cabinet; control the config-port drop, serial terminals, and the module; config drift is visible via PCB compare / AssetCentre. **Ext. tools:** Physical inspection (cabinet/seal, config-port, serial); PCB/AssetCentre compare.	Chassis secured; interfaces controlled; drift detectable.	EDR-specific.
EDR 3.12	Provisioning product-supplier roots of trust	**N/A.** No documented signed-firmware supplier root of trust. Record N/A; compensate with firmware provenance (3.10), config-port lockdown, physical control, and segmentation. **Ext. tools:** Confirm no RoT; ProSoft provenance process.	No supplier RoT; compensating controls recorded.	N/A.
EDR 3.13	Provisioning asset-owner roots of trust	**N/A.** No asset-owner certificate provisioning (no PKI). Record N/A; asset-owner trust at the network + physical + processor layer. **Ext. tools:** Confirm no cert provisioning; segmentation review.	No asset-owner RoT; compensating controls recorded.	N/A.
EDR 3.14	Integrity of the boot process	**N/A.** No verified/secure boot. Record N/A; mitigate via firmware provenance (3.10), config-port lockdown, physical control, and monitoring. **Ext. tools:** Confirm no secure-boot capability.	No verified boot; compensating controls recorded.	N/A.

FR 4 — Data Confidentiality (DC)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 4.1	Information confidentiality	Confirm information confidentiality — sensitive data is protected in transit and at rest. **Steps:** for this product this is assessed with the same evidence as **SR/CR 3.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Information confidentiality verified individually.	
SR/CR 4.2	Information persistence	On decommission/repurpose, clear the module config (download a default/blank .ppf) and destroy/secure the **.ppf** + backups so the Modbus maps, addresses, and command lists are not recoverable. **Ext. tools:** PCB (download default/blank); .ppf + backup sanitize.	Module config + maps purged on decommission or reuse.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 4.3	Use of cryptography	**N/A.** No comms crypto on the module. Rely on segmentation; where confidentiality is required, add an external secure layer. **Ext. tools:** Confirm no crypto; segmentation review.	No comms crypto - compensating segmentation.	N/A.

FR 5 — Restricted Data Flow (RDF)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 5.1	Network segmentation	The module **bridges the ControlLogix backplane to a serial Modbus network** - treat it as a conduit: keep the serial loop contained, and put the **config Ethernet port** on a controlled config/OT segment (or disable it). Only required flows cross; no untrusted reach to the config port. **Ext. tools:** `nmap -p 80` per zone; serial loop + config-net segmentation review.	Serial loop contained; config port segmented/disabled; module conduit controlled.	
SR 5.2	Zone boundary protection	Confirm zone-boundary protection — only required flows cross the boundary (system SR / network device). **Steps:** for this product this is assessed with the same evidence as **SR/CR 5.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Zone boundary protection verified individually.	
SR 5.3	General-purpose person-to-person comms restrictions	**N/A.** Single-purpose comms module. **Ext. tools:** Confirm no such services.	N/A - justification recorded.	N/A
SR 5.4	Application partitioning	Confirm the **config/diagnostics path** is separated from the operational serial Modbus side, and the module cleanly partitions the ControlLogix (backplane) from the serial Modbus network. **Ext. tools:** Config-net vs serial-side separation review.	Config and operational sides separated; backplane/serial partitioned.	

FR 6 — Timely Response to Events (TRE)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 6.1	Audit log accessibility	No native module log - monitor off-module: the **ControlLogix processor + AssetCentre** for config/AOI changes, the module **LED/web diagnostics** + AOI status for health, and a serial/Modbus-aware **OT IDS** on the loop; forward to the SIEM. **Ext. tools:** AssetCentre; module diagnostics/AOI status; serial/Modbus IDS + SIEM.	Module/serial events monitored off-module in near-real-time.	No native log - Compensating.
SR/CR 6.2	Continuous monitoring	Confirm continuous monitoring — events forwarded to and monitored by a SIEM/OT-IDS in near-real-time. **Steps:** for this product this is assessed with the same evidence as **SR/CR 6.1** above (shared steps); confirm this requirement individually and adjudicate it separately.	Continuous monitoring verified individually.	

FR 7 — Resource Availability (RA)

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 7.1	Denial-of-service protection	Confirm resilience: a serial-Modbus storm or config-port flood must not stop the backplane data exchange or drive unsafe process action - verify the comm-fail behavior is safe; rate-limit/segment the config port; disable it when unused. **Ext. tools:** ISOLATED bench: serial-Modbus storm; confirm safe comm-fail + backplane continuity.	Backplane exchange continues (or fails safe) under comms load; config port protected.	
SR/CR 7.2	Resource management	Confirm bounded resources: per-port command limits (up to 325/port), the 10,000-register map, and connection limits so one service cannot starve the exchange. **Ext. tools:** PCB (command/register/connection config); bench load test.	Resources bounded; no exhaustion-based loss of the exchange.	

Req (3-3 / 4-2)	Requirement	Where & how to verify (native steps + Ext. tools)	Pass criteria / evidence	Notes / lifecycle
SR/CR 7.3	Control system backup	Confirm the PCB **.ppf** + the Studio 5000 project (AOI/tags) + firmware record are backed up (versioned, offline) and a restore is tested on a spare module for a known-good rebuild. **Ext. tools:** Backup repository; `certutil -hashfile .ppf SHA256`; restore test (spare module).	Backups current and recovery validated.	
SR/CR 7.4	Control system recovery & reconstitution	Confirm recovery/reconstitution — a restore to a known-good state is tested. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.3** above (shared steps); confirm this requirement individually and adjudicate it separately.	Control system recovery & reconstitution verified individually.	
SR/CR 7.5	Emergency Power	**N/A.** Emergency/backup power is provided at the facility/UPS level, not the device; confirm the device retains its security state across a power transfer.	N/A - justification recorded (not assessed as a control).	N/A
SR/CR 7.6	Network & security configuration settings	Harden: **disable the config Ethernet port + web after commissioning**, use only the required serial ports/protocol (RTU or ASCII; Master or Slave), restrict writable registers/commands, and patch firmware; the ControlLogix keyswitch/FactoryTalk govern the processor side. **Ext. tools:** PCB (ports/commands); confirm config port disabled; `nmap -p 80` (off/segmented).	Least functionality; config port/web disabled; required serial only; writes scoped; patched.	
SR/CR 7.7	Least functionality	Confirm least functionality — unused services/ports/protocols are disabled. **Steps:** for this product this is assessed with the same evidence as **SR/CR 7.6** above (shared steps); confirm this requirement individually and adjudicate it separately.	Least functionality verified individually.	
SR/CR 7.8	Component inventory	Inventory the module (MVI56E-MCM / MCMXT + firmware rev), its slot/chassis, and the serial config, so vulnerabilities track against ProSoft / CISA advisories. **Ext. tools:** PCB / module LED (model/firmware); Studio 5000 (slot); asset register; ProSoft advisory check.	Accurate module/firmware inventory maintained; advisories tracked.	

External Test Tools — how to use

Tool / method	Verifies	How to use — steps / example	Constraints
ProSoft Configuration Builder (PCB)	Native config, command-list & diagnostics (most FRs)	Review the module config (serial ports, Master/Slave, command list, register map), diagnostics, and firmware; back up the .ppf. Connect over the config Ethernet port or via CIPconnect. Disable the config port after commissioning.	Read/authorized project copy.
Studio 5000 AOI + FactoryTalk Security/AssetCentre	Processor-side authz & audit (1.1, 2.1, 2.8, 3.4)	The MVI56E-MCM is owned by the ControlLogix processor - review the AOI/controller tags, FactoryTalk Security roles, keyswitch, and AssetCentre change records.	Read-only; authorized.
PDS / on-board web / curl	Config-port exposure check (1.1, 1.13, 7.6)	ProSoft Discovery Service + `curl -i http:///`; confirm the config port + web are segmented/disabled after commissioning (docs only, no config exposure).	Read-only; authorized.
modpoll / mbpoll (serial)	Serial Modbus read-back & write-scope (1.2, 2.1, 3.3)	`modpoll -m rtu -a -b -1 /dev/ttyUSB0`; read status; confirm which registers/commands are writable (authorized/isolated).	Read-only on production; writes isolated.
nmap	Config-port exposure & segmentation (1.13, 5.x, 7.6)	`nmap -Pn -p 80` ; confirm the config Ethernet port/web is reachable only from the authorized segment (no Modbus TCP on MCM).	Read-only; authorized.
Serial analyzer / Wireshark	Comms integrity & containment (1.2, 3.1)	Capture the serial Modbus (USB-RS485 tap) or the config-port traffic; confirm cleartext stays contained; inspect writes.	Passive; tap/SPAN only.
Serial/Modbus fuzzer (isolated)	Input validation (3.5)	`boofuzz` serial Modbus / config to a BENCH module; watch the LED + AOI status for faulting.	ISOLATED bench; written authorization.
Serial/Modbus OT IDS + ProSoft/CISA advisories	Monitoring & vuln tracking (2.8, 6.x, 7.8)	IDS on the serial loop/config net for anomalous writes; track ProSoft / CISA advisories + required firmware.	Passive; coordinate with SOC.

Controlled - Project Use · Independent Assurance Services · ISA/IEC 62443-3-3 / 4-2 · Product-specific companion to the 62443 assessment tool; confirm exact menu paths / CLI against the deployed firmware and the project hardening baseline. ISOLATED tools run only in a representative/lab environment with written authorization.